Design and Implementation of 4-Bit R-2R Ladder DAC using LM358

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Motivation and Problem Statement

- In electronics, circuits built on breadboards are often unstable and not suitable for long-term use. To improve reliability and performance, circuits are designed on PCBs. The motivation of this project is to gain practical knowledge of PCB design, fabrication, and testing using an op-amp (LM358).

- The problem is to design and implement an op-amp-based circuit on a PCB that can process input signals effectively. The circuit is tested using a function generator as input and an oscilloscope to observe the output. The goal is to ensure stable operation and verify that the output matches the expected behavior.

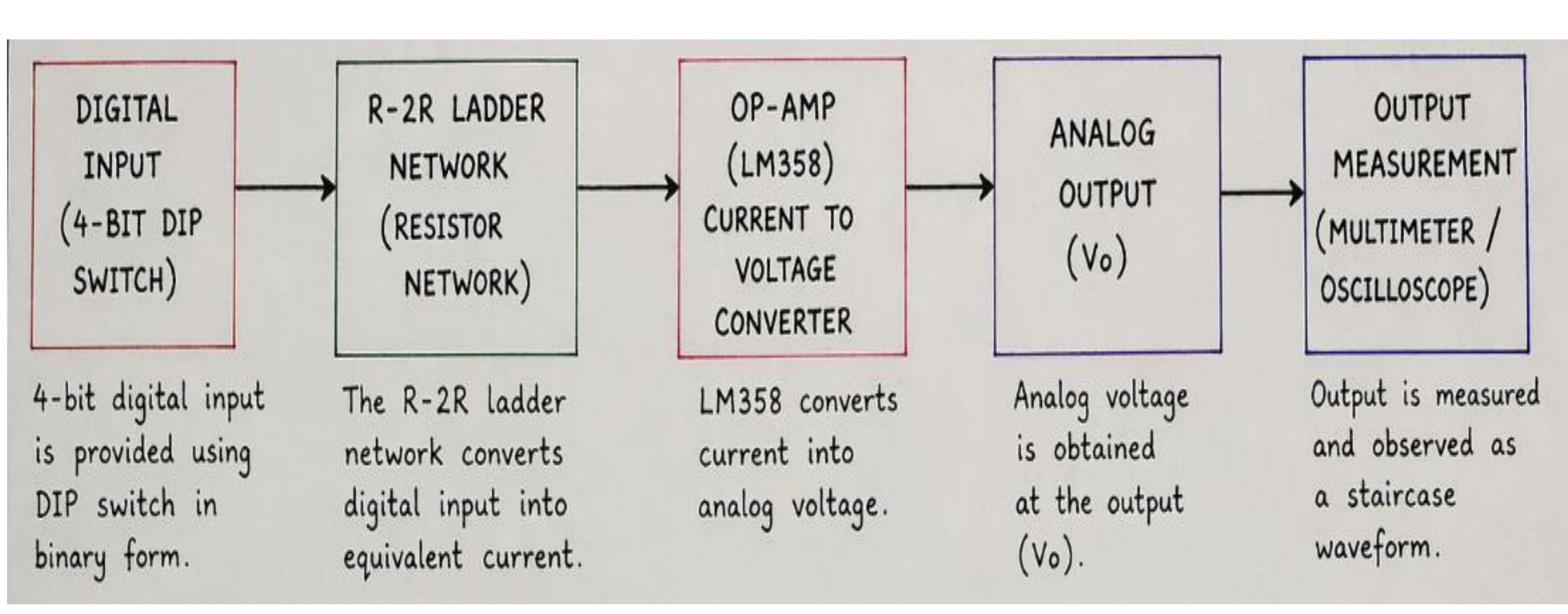
Proposed Solution

- We propose a simple method to convert digital signals into analog voltage using a 4-bit R-2R Ladder DAC circuit.

- A DIP switch provides 4-bit digital input (0000–1111), which is applied to the R-2R resistor network. This network converts the binary input into current. The LM358 op-amp then converts this current into analog voltage.

- The output is taken from the VO pin and varies step-by-step according to the input, forming a staircase waveform.

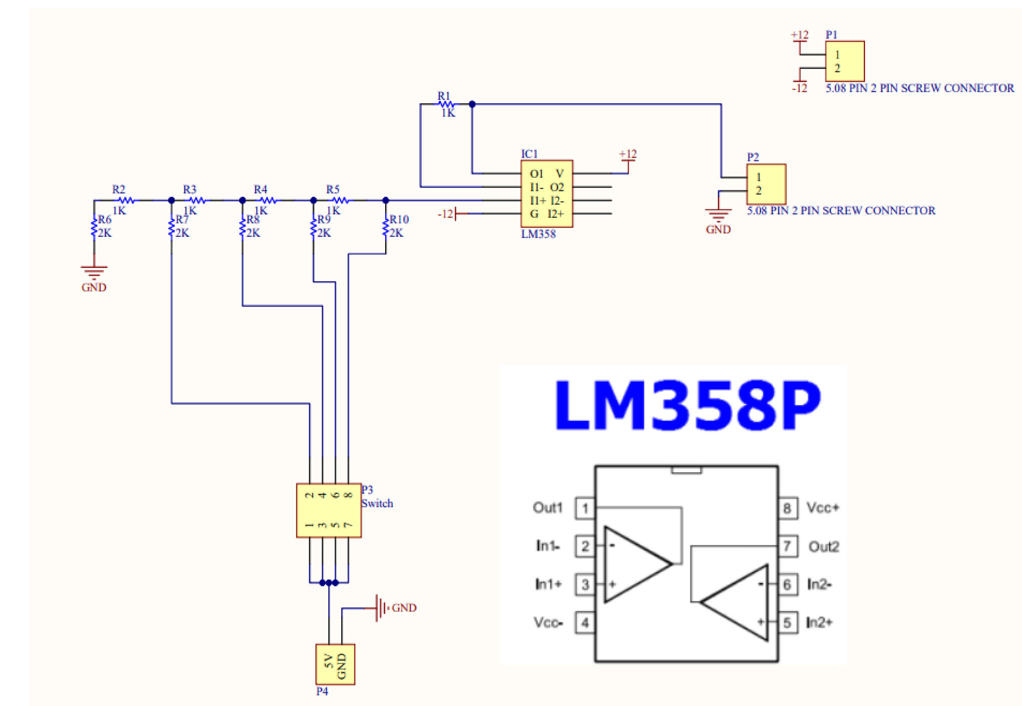
- The circuit is designed, converted into a PCB layout, and fabricated using Gerber files. After fabrication, components are assembled and tested. The output is measured using a multimeter or oscilloscope.



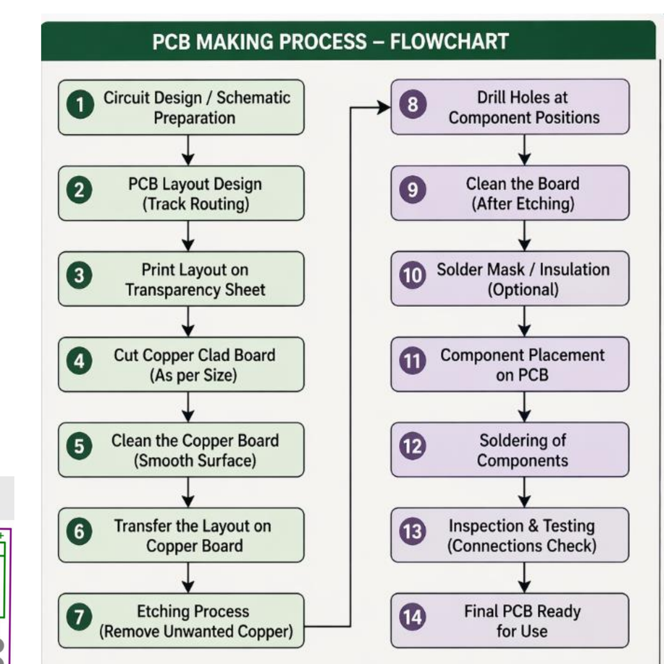
Implementation and Results

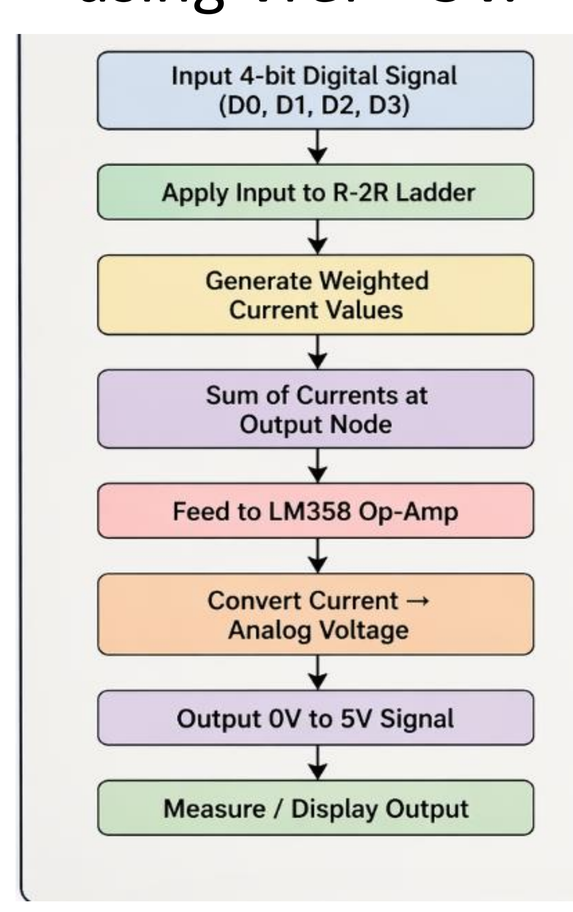
- A 4-bit input (D0–D3) is given to an R-2R ladder (1 kΩ, 2.2 kΩ), which converts it into weighted

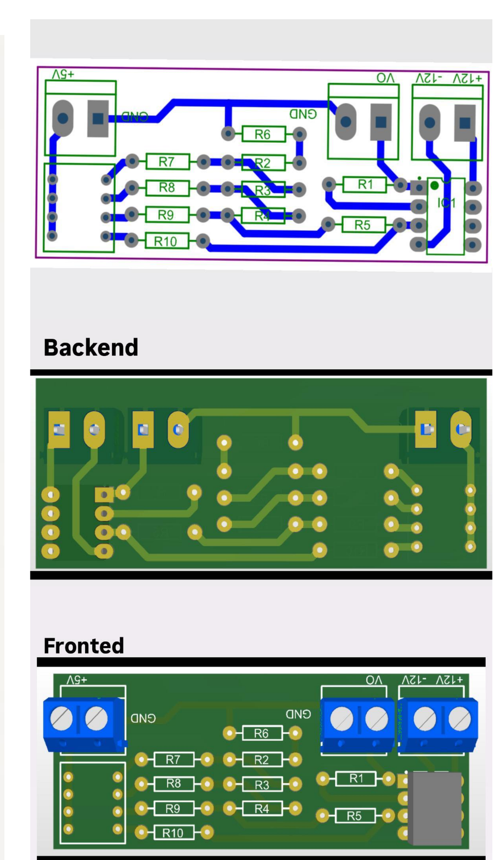
$$V_{out} = V_{ref} \times \frac{D}{2^n}$$



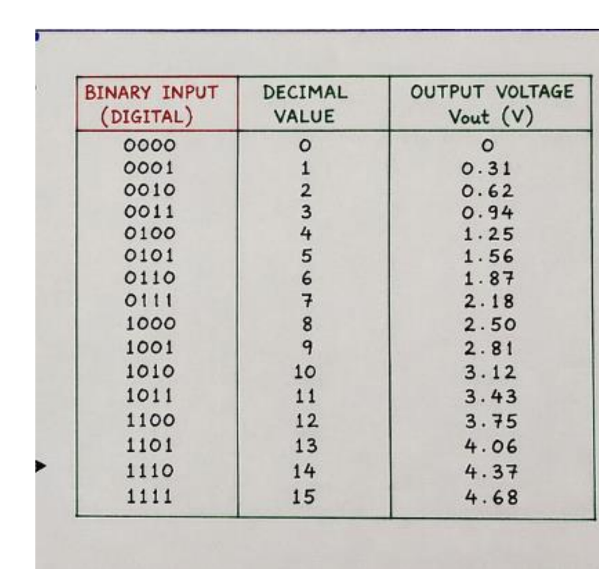
- These currents are summed at the output node and fed to an LM358 op-amp (±12V supply), which converts them into an analog voltage using Vref = 5V.





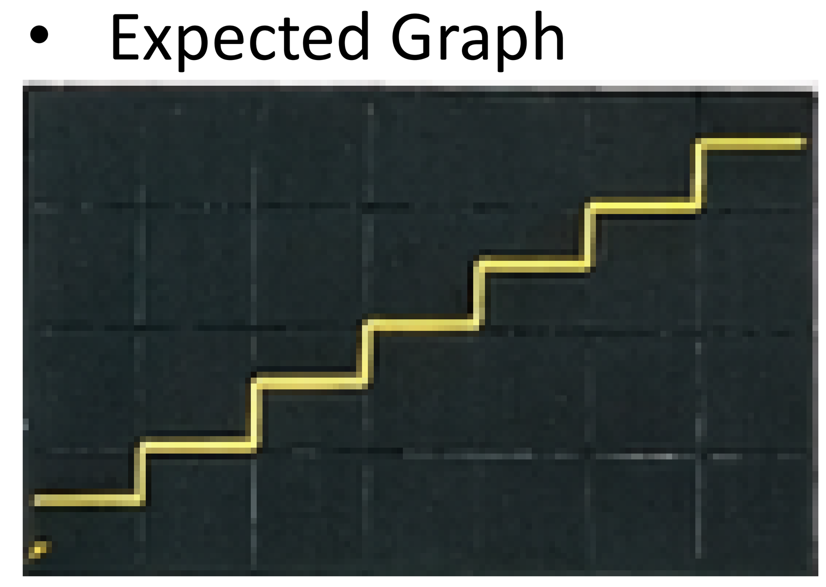


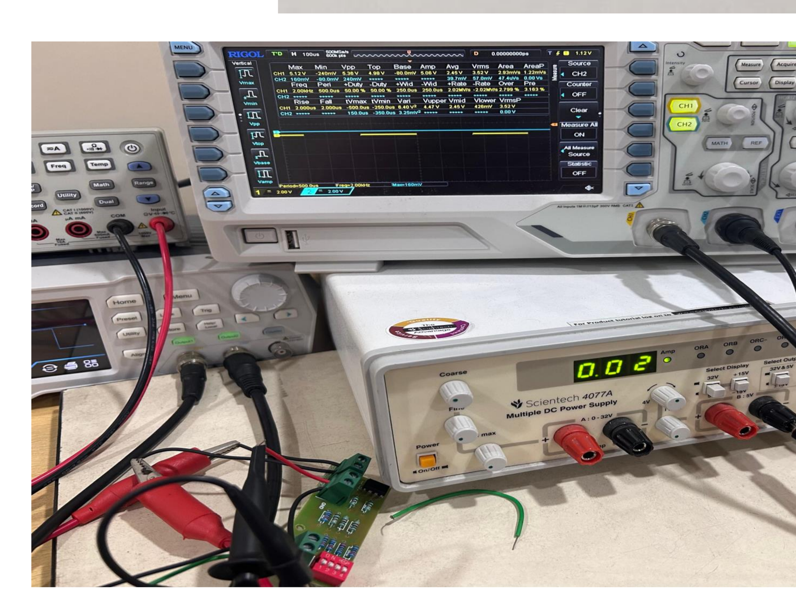
- Calculated Result



BINARY INPUT (DIGITAL)	DECIMAL VALUE	OUTPUT VOLTAGE Vout (V)
0000	0	0
0001	1	0.31
0010	2	0.62
0011	3	0.94
0100	4	1.25
0101	5	1.56
0110	6	1.87
0111	7	2.18
1000	8	2.50
1001	9	2.81
1010	10	3.12
1011	11	3.43
1100	12	3.75
1101	13	4.06
1110	14	4.37
1111	15	4.68

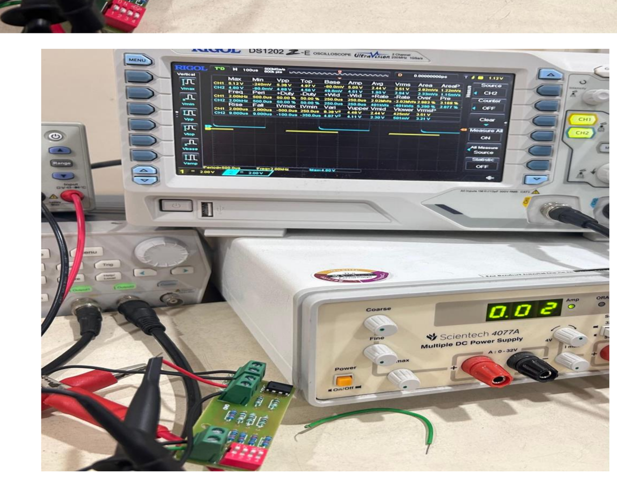
- Expected Graph





- Practical Result

At input 0000, output is 0V and at 1111, output is approximately 4.80V.



Conclusion

The R-2R DAC gives step-wise analog output, but the full 0–5V range is limited by resistor tolerance, LM358, and loading effects. This can be improved using an Arduino.